

Single-Seed Benchmarks Are Unreliable for LLM Coding Agents on Bioinformatics Tasks

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Abstract

Large language model (LLM) coding agents are increasingly applied to bioinformatics analysis—differential expression, single-cell clustering, survival modeling, rare-variant prioritization—and are compared on benchmarks that score a *single run* per configuration, graded by an LLM judge. We show that such single-seed benchmarks are unreliable for ranking these agents. On 126 pre-registered runs spanning six bioinformatics tasks, three tool configurations, and seven random seeds per cell, seed-to-seed variability dominates and swamps the configuration differences the benchmark is meant to detect: a generalizability-theory decomposition attributes 51% of score variance to task and 0% to configuration at the point estimate, and no pairwise configuration comparison is significant (0/18). The configuration reliability coefficient is *unidentified*—a clamped point estimate of zero flanked by intervals spanning essentially the whole scale ($E\rho_{\text{config}}^2$: analysis-of-variance (ANOVA) point 0.00; nonparametric bootstrap 95% confidence interval (CI) [0.00, 0.81]; Bayesian half-Cauchy posterior median 0.46, 95% credible interval [0.00, 0.99])—so we claim unidentifiability, not the absence of an effect. Most concretely, the configuration leaderboard is fragile: resampling seeds leaves the observed best configuration on top only 42% of the time (chance 33%; 37% under task-and-seed resampling), so a ranking read from a single-seed run is close to a coin flip. A second, downstream noise source is the LLM judge: scoring is 6% non-deterministic per call at temperature 0 and up to 10% order-sensitive, and documentation/justification criteria are where two judges diverge most—**gpt-4o-mini** is systematically stricter (passing 39% of such items) than **gpt-4o** (which passes all), a calibration offset that drives chance-corrected agreement to zero ($\kappa_c = 0.00$) while capability criteria reach $\kappa_a = 0.87$; the capability-to-documentation *drop* replicates against a cross-family Llama-3.3-70B judge ($\kappa_a = 0.87 \rightarrow \kappa_c = 0.23$). For bioinformatics agent evaluation, where a benchmark is meant to guide real methodological choices, we argue that multi-seed reliability—not single-run accuracy—is a prerequisite for trustworthy comparison, and we distill a *decompose-before-you-rank* protocol to that end.

1 Introduction

Bioinformatics is fast becoming a testbed for LLM coding agents, but the benchmarks used to compare them are scored from a single run—and a single run is not a reliable measurement. Agents built on models such as GPT-4 and Claude are now evaluated on computational-biology tasks—differential-expression analysis, single-cell clustering and integration, survival modeling, rare-variant prioritization—by benchmarks such as BixBench (Mitchener et al., 2025) and GenoTEX, typically reporting one graded score per system and ranking accordingly. But a score is a measurement, and a measurement without a characterized noise floor is uninterpretable. If seed-to-seed variability—the run-to-run stochasticity of a temperature-sampled agent solving the same task—is larger than the differences a benchmark reports between systems, then its rankings are noise dressed as signal. Despite a mature literature on seed variance in machine learning (Reimers and Gurevych, 2017; Bouthillier et al., 2021; Bethard, 2022; Henderson et al., 2018) and on LLM-judge bias (Wang et al., 2023; Zheng et al., 2023), the specific question—how much of a bioinformatics *agent* benchmark’s score variance is real signal about the systems it compares—has not been answered with the tools of measurement theory.

We answer it for one deeply instrumented bioinformatics agent benchmark. Rather than add breadth (more tasks, more models), we add depth: 126 pre-registered runs spanning six bioinformatics tasks, three tool configurations (**bare**, **nia**, **full**), and seven seeds per cell, graded by a modal-of-3 LLM judge whose protocol, thresholds, and analysis plan were committed to a version-controlled repository before any score was inspected. This is a reliability study, not a new leaderboard: we use the benchmark to show what characterizing an agent benchmark’s noise floor reveals and requires. On this frozen data we run a generalizability-theory (G-theory) variance decomposition (Cronbach et al., 1972; Brennan, 2001), quantify three downstream judge-noise sources, and characterize cross-judge and cross-family agreement stratified by criterion type.

Our contributions are:

1. **Single-seed rankings on this bioinformatics benchmark are unreliable.** Seed variance dominates: configuration explains 0% of score variance at the point estimate, no pairwise comparison is significant (0/18), and three uncertainty methods agree the configuration reliability coefficient cannot be pinned down (its 95% interval spans nearly $[0, 1]$). Under seed resampling the best configuration stays best only 42% of the time (chance 33%), so a single-run leaderboard is close to a coin flip. We convert a null result into a positive, decision-theoretic statement about ranking risk, while being explicit that we show unidentifiability, not the absence of an effect.
2. **Documentation criteria are where LLM judges are least reproducible.** Beyond 6% per-call non-determinism and $\leq 10\%$ artifact-order sensitivity, two same-family judges diverge sharply on documentation/justification criteria—a systematic strictness offset that forces chance-corrected agreement to zero ($\kappa_c = 0.00$) where capability criteria reach $\kappa_a = 0.87$ —and the capability-to-documentation drop replicates against a cross-family judge. We disclose the mechanism (one judge is uniformly more lenient) and anchor it, tentatively, against human labels.
3. **A reliability protocol for bioinformatics agent benchmarks.** We distill a *decompose-before-you-rank* discipline—pre-register the judge protocol, run a G-theory decomposition over {task, system, seed, judge}, compute the reliability coefficient for the factor being ranked, and refuse to publish a ranking when it is unidentified—plus a minimal human-validity anchor. For a computational-biology audience the practical upshot is concrete: report multi-seed reliability, not single-run accuracy, before drawing methodological conclusions from an agent benchmark.

2 Related Work

Our work sits at the intersection of LLM-judge reliability, agent benchmarking, and the measurement-theoretic turn in AI evaluation.

LLM-as-judge reliability. The judge paradigm was consolidated by Zheng et al. (2023) and Dubois et al. (2023), who argued LLM–human agreement rivals inter-human agreement. A subsequent literature documented systematic failure modes: order/position bias (Wang et al., 2023), cognitive biases (Koo et al., 2023), self-preference (Panickssery et al., 2024), and low inter-sample agreement (Stureborg et al., 2024). Mitigations include judge panels (Verga et al., 2024) and per-property calibration against humans (Bavaresco et al., 2024). Norman et al. (2026) and Dietz et al. (2025) recently formalized that judges can be highly *reliable* yet *invalid*, motivating a minimum validation protocol. The closest precedent on judge non-determinism is Atil et al. (2024), who measured LLM-as-solver variability at temperature 0; we measure the analogous quantity for the LLM *as judge*, and we extend the candidate-order bias of Wang et al. (2023) to a within-artifact section-order permutation.

Agent benchmarking and its rigor. The coding-agent template was set by Jimenez et al. (2024); Yang et al. (2024), with broad suites following (Liu et al., 2024; Mialon et al., 2024; Ma et al., 2024). Kapoor et al. (2024) and Kapoor et al. (2025) argued for cost-aware, standardized, distributional evaluation. In bioinformatics specifically, Mitchener et al. (2025) is the nearest domain peer, and Siegel et al. (2024) asks whether agents can reproduce research—we ask the parallel question of whether agent benchmarks themselves are reproducible, grounding task design in computational-reproducibility norms (Sandve et al., 2013). A rigor literature has recently emerged: Reuel et al. (2024) and Zhu et al. (2025) ship benchmark-quality checklists, and Miller (2024) supplies error-bar conventions for evals; our protocol specializes these to the reliability of the object of comparison.

Variance decomposition and measurement validity. Generalizability theory (Cronbach et al., 1972; Brennan, 2001) decomposes a measurement’s variance into facet components and defines reliability as the share attributable to the object of measurement. It is now being imported into LLM evaluation: Song et al. (2025) apply G-theory to LLM autoscoring, and, most closely, Messing (2026) integrate total-survey-error and G-theory into a “Total Evaluation Error” framework for LLM pipelines. We differentiate from this methodological neighbor in three ways: our setting is agent *tool configurations* rather than annotation pipelines; our headline is a *construct-specific* judge collapse ($\kappa_c = 0$) rather than aggregate error; and we show *cross-family* replication. On validity, Jacobs and Wallach (2021) and Raji et al. (2021) frame construct validity for AI measurement, and Madaan et al. (2025) quantify seed variance in evaluation benchmarks. That analytical-flexibility variance is a structural feature of science is established by many-analyst studies that obtain divergent conclusions on identical data (Botvinik-Nezer et al., 2020; Schweinsberg et al., 2021); our seed-variance result is the LLM-agent analogue. Our decision-theoretic reanalysis draws on the ranking (in)stability literature (Boubdir et al., 2024; Singh et al., 2025; Huang et al., 2025; Chiang et al., 2024) and the statistical-power framing of Card et al. (2020).

3 Methods

Every scoring and analysis decision was locked in version control before any score was inspected. We summarize the design; full pre-registration artifacts, commit hashes, and data checksums are in the supplement.

Tasks and configurations. We evaluate a six-task pilot drawn from a 14-task bioinformatics registry so that each of three difficulty tiers contributes two tasks and contested-decision tasks sit alongside tasks with unambiguous ground truth: bulk RNA-seq differential expression (T1.1), functional enrichment (T1.4), single-cell clustering (T2.1), multivariable Cox regression (T2.4), single-cell integration (T3.1), and rare-variant prioritization (T3.4). All runs use Anthropic `claude-sonnet-4-6` through the Claude Code CLI under three configurations differing only in tool-augmentation surface: `bare` (built-in tools only), `nia` (a curated, internet-free MCP tool set), and `full` (an added plugin). All three share an identical per-run budget cap and setting-source restriction, so the configuration is the only manipulated variable. The full $6 \times 3 \times 7 = 126$ -run design is balanced and complete.

Judge protocol. For each (run, criterion) pair, a `ModalJudge(GPT4oJudge(gpt-4o-mini))` chain issued three independent temperature-0 calls and returned the mode; ties among rubric items triggered a fourth call. The choice of modal-of-3 was empirical: a 100-call determinism study (§5) found single-call scoring too non-deterministic to defend as a per-item measurement. Each

run yields a composite checklist score `c_score` (fraction of checklist criteria passed, with contested criteria contributing through a normalized 0–2 rubric), the primary dependent variable throughout.

Pre-registration. Blessing thresholds ($\geq 95\%$ per-item modal agreement between independent scoring sessions), the calibration hypothesis ($\kappa_a > \kappa_c + 0.1$), the criterion-behavior classification schema, and the judge-by-alias policy were each committed before scoring. We adopt pre-registration precisely because the failure modes we anticipated—judge non-determinism, alias drift, class-conditional disagreement—are easy to rationalize post hoc (Forde and Paganini, 2019; van Miltenburg et al., 2021).

Statistical analysis. Configuration effects were tested by stratified two-sided permutation tests (10,000 permutations) across the 18 (task, config-pair) comparisons, with Benjamini–Hochberg (BH) false-discovery-rate correction (Benjamini and Hochberg, 1995). We decompose `c_score` variance by G-theory (§4), attaching uncertainty (all intervals are 95% confidence intervals, CI, unless noted) three ways: a nonparametric design bootstrap that resamples tasks and seeds and retains the *signed* variance estimator so the interval has real spread (Tong and Brennan, 2007; Searle et al., 1992; Gelman, 2005); a Bayesian crossed random-effects model with half-Cauchy priors on all standard deviations (Gelman, 2006; LoPilato et al., 2015); and a restricted-maximum-likelihood (REML) cross-check. We add a ranking-reversal reanalysis, two one-sided equivalence tests (TOST) (Lakens, 2017; Lakens et al., 2018) (a Bayesian region-of-practical-equivalence check (Kruschke, 2018) would serve the same role), and—for judge validity—exact Clopper–Pearson (Clopper and Pearson, 1934) and bootstrap- κ (Klar et al., 2002) intervals. All code is deterministic (`numpy default_rng(42)`).

4 Seed Variance Swamps Configuration Effects

The benchmark cannot identify the tool configurations it is designed to compare—not merely “no significant difference,” but a reliability coefficient no method can pin down (its interval spans nearly the whole scale) and a leaderboard that reverses under trivial resampling. The core picture is visible directly in the raw scores (Figure 1): within every task the spread of `c_score` across the seven seeds is large relative to the gap between the three configuration means, so which configuration “wins” on any single seed is largely a draw from that spread.

No configuration effect is detectable. Within-cell seed variability exceeds every between-configuration mean difference: the median within-cell standard deviation of `c_score` is 0.135 (max 0.319), while the largest of 18 pairwise configuration differences is $|\Delta| = 0.143$. No comparison is significant after correction (0/18 at $p_{\text{BH}} < 0.05$; smallest $p_{\text{raw}} = 0.076$, smallest $p_{\text{BH}} = 0.877$). A power analysis shows this is structural: at $n = 7$ seeds and the median observed σ , detecting an effect at 80% power requires $|\Delta| \approx 0.20$, and 0/18 observed differences reach that threshold. A forest plot of all 18 pairwise differences with bootstrap confidence intervals (Figure 2) shows 16 of 18 intervals straddling zero; the two that do not—T1.1 **bare**–full and T3.1 **full**–nia, the largest observed $|\Delta| = 0.143$ —are exactly the two smallest- p comparisons ($p_{\text{raw}} = 0.076, 0.098$), and neither survives multiple-comparison correction, so we do not read them as real effects.

Variance decomposition. The G-theory decomposition (Figure 3) attributes 51.1% of `c_score` variance to task, 41.4% to residual (the three-way task $\times$ config $\times$ seed term), 6.0% to task $\times$ seed, and 1.6% to the config $\times$ seed and task $\times$ config two-way terms combined; configuration and seed

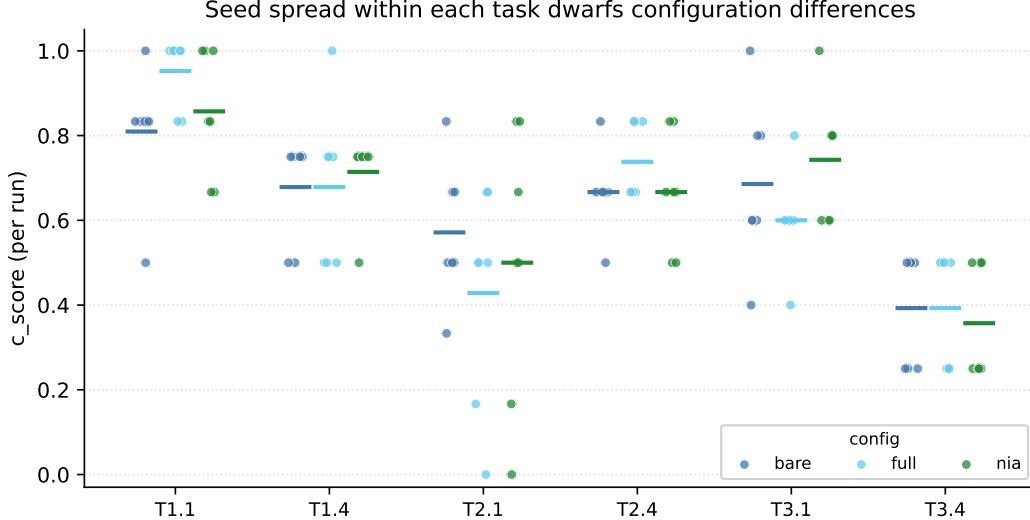


Figure 1: A single seed is not a reliable measurement. Every one of the 126 runs, one column per bioinformatics task, points colored by tool configuration (**bare**/**full**/**nia**), horizontal bars are per-configuration means. Within each task the seed-to-seed spread is comparable to or larger than the distance between configuration means, so a single-run comparison of configurations is dominated by seed noise.

main effects are both 0.0% (these shares are computed on positive-clamped components, so the two retained large shares are mildly upward-biased; the signed-bootstrap intervals in Table 1 are the honest object). The configuration generalizability coefficient at the realized design is $E\rho_{\text{config}}^2 = 0.00$, whereas the benchmark separates *tasks* with a high coefficient, $E\rho_{\text{task}}^2 = 0.94$. Two readings of this decomposition should be kept in view. First, because **c_score** is judge-graded, the residual term bounds *both* agent seed-to-seed stochasticity and any residual judge noise (§5) that modal-of-3 does not remove; it is an upper bound on agent-only variability, not a pure estimate of it. Second, the large task share partly reflects structural differences in score range across tasks (distinct rubrics, ceiling **j_norm** on some tasks, and the near-degenerate support of T3.4), so $E\rho_{\text{task}}^2$ is best read as a sanity check that the instrument is not pure noise—it can separate *something* reliably—rather than as a benchmark virtue. The finding of interest is the contrast: the facet the benchmark targets is the one it cannot resolve.

The zero is unidentified, not proven. A variance component estimated at zero by analysis of variance (ANOVA) is a boundary artifact of negative-component clamping, so we do not claim configuration truly has no effect. Instead, three uncertainty methods agree the coefficient is *unidentifiable* at this scale (Table 2): the design bootstrap gives $E\rho_{\text{config}}^2$ 95% CI [0.00, 0.81] (configuration variance share 0.0%, CI [0.0, 9.6]%), and a prior-robust Bayesian posterior gives median 0.46 with 95% credible interval [0.00, 0.99]. We stress that the Bayesian central estimate is *not* near zero—at only three configuration levels the config SD is prior-driven and its posterior is wide—so the honest summary is unidentifiability (an interval spanning almost [0, 1]), not “configuration reliability is zero.” What all three methods concur on is only the negative: none can establish that configurations are reliably separable. Task reliability, by contrast, is comparatively well-identified ($E\rho_{\text{task}}^2$ bootstrap [0.21, 0.98], Bayesian [0.82, 0.99]), subject to the score-range caveat above.

Table 1: Variance components of `c_score` (% of total), with signed nonparametric bootstrap 95% CIs. Point estimates are the ANOVA method-of-moments components (positive-clamped); intervals come from resampling the 6 tasks and 7 seeds (10,000 replicates, signed estimator). Configuration and seed main effects are estimated at zero, with upper bounds of 9.6% and 12.7% respectively; task and residual dominate.

Component	% of total variance	95% CI
task	51.1	[2.7, 74.4]
residual (task×config×seed)	41.4	[12.8, 52.3]
task×seed	6.0	[0.0, 24.8]
config×seed	0.9	[0.0, 33.9]
task×config	0.7	[0.0, 24.2]
config (main)	0.0	[0.0, 9.6]
seed (main)	0.0	[0.0, 12.7]

Table 2: Configuration reliability is unidentified; task reliability is comparatively identified. Three uncertainty methods on the same 126 runs. The generalizability coefficient $E\rho^2$ for configuration has a lower bound at zero and an upper bound near one, and its Bayesian central estimate (0.46) is itself prior-driven—the design does not identify it. Task separability is high but partly reflects deliberate score-range differences across tasks (see text).

Method	$E\rho_{\text{config}}^2$	$E\rho_{\text{task}}^2$
ANOVA point estimate (G-study)	0.00	0.94
Design bootstrap, 95% CI	0.00 [0.00, 0.81]	0.94 [0.21, 0.98]
Bayesian half-Cauchy, 95% CrI	0.46 [0.00, 0.99]	0.95 [0.82, 0.99]

The leaderboard is unstable. We make the unidentifiability concrete by resampling and rebuilding the configuration leaderboard 10,000 times (Figure 4). Under task-and-seed resampling the observed best configuration remains best in only 37.1% of resamples—barely above the 33.3% chance rate—any pairwise reversal occurs in 76.7%, and the mean Kendall τ against the observed order is 0.07. Because task variance dominates, most of this churn is task-driven; a seed-only resample (tasks held fixed) is the honest test of the seed-limited claim, and there the top configuration still changes over half the time (42.1% unchanged). The ordering is not, however, perfectly symmetric: the tool-augmented configurations top the leaderboard more often than **bare** (**nia** 37.1%, **full** 36.4%, **bare** 26.5%), a weak hint that tools may help. That hint does not survive testing—no **bare**-vs-augmented comparison is significant, and pooling **bare** against **{nia,full}** does not change the picture—so we report it as an unresolved tendency the design cannot adjudicate, not as noise pretending to be signal (Huang et al., 2025; Boubdir et al., 2024).

The design is uninformative at the effect sizes that matter. A decision study is conditional on the very quantity that is unidentified: *if* the configuration variance component is near its point estimate, no number of seeds raises $E\rho_{\text{config}}^2$ (reliability is effect-limited, not seed-limited); *if* it is near the Bayesian median, a modest number of additional seeds and tasks would identify it. The design cannot distinguish these cases—which is the point. Complementary equivalence tests (TOST) make the scope explicit: at a smallest-effect-of-interest (SESOI) of 0.10 `c_score` units—roughly one checklist criterion out of ten, a defensible floor for a “meaningful” tool effect, reported here as one point in a sensitivity sweep over $\{0.05, 0.10, 0.15, 0.20\}$ —0/18 comparisons are statistically equivalent and 0/18 are significant: the design can neither detect nor rule out effects

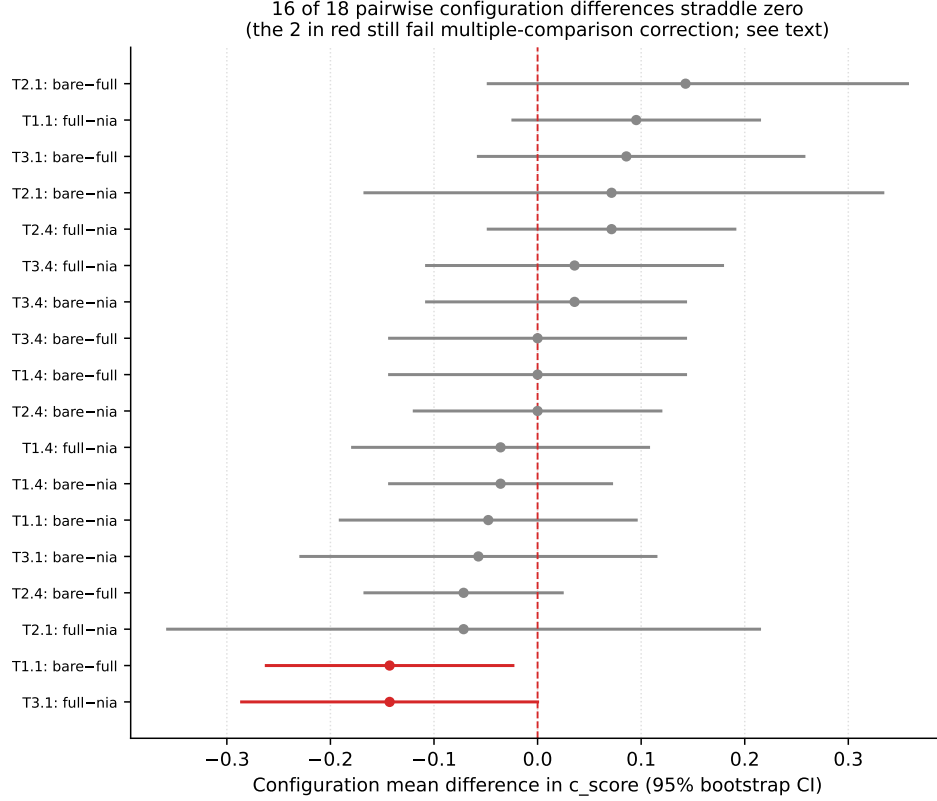


Figure 2: No configuration comparison is cleanly separated from zero. All 18 pairwise (task, configuration-pair) mean differences in `c_score` with percentile bootstrap 95% CIs (resampling the 7 seeds per group). 16 of 18 intervals straddle zero (grey); the 2 that do not (red) are the largest observed differences but do not survive Benjamini–Hochberg correction (§4), so no configuration effect is established.

in the practically relevant range. Equivalence is reached for 3/18 comparisons at a SESOI of 0.15 and 10/18 at a SESOI of 0.20.

5 The Judge Is a Distinct Noise Source, and Documentation Criteria Are Where Judges Diverge

The judge is a distinct, irreducible source of measurement noise—not the whole residual, which our design cannot partition between agent and judge, but a characterized component whose sharpest failure is a systematic disagreement between judges on documentation-style criteria. Three judge-noise sources sit downstream of the agent and compound rather than substitute; we quantify each and then localize the largest between-judge divergence to a specific criterion type.

Intra-session non-determinism. Re-scoring 20 stratified items five times each (100 calls) at temperature 0 yields a 6.0% per-call flip rate (Wilson 95% CI 2.8–12.5%), concentrated on the borderline-decision task T2.1 (11% vs. 0% on T1.1). This is the LLM-as-judge analogue of the solver-side non-determinism of Atil et al. (2024); modal-of-3 attenuates it to a ~ 1 –2% inter-session floor, which we adopt for all scoring.

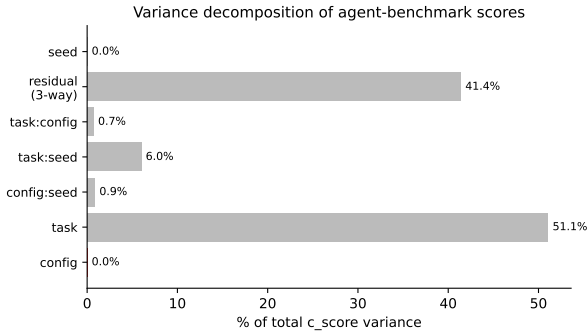


Figure 3: Task explains 51% of score variance; configuration explains 0%. G-theory variance decomposition of `c_score` across 126 runs. The factor the benchmark is built to compare (configuration, red) contributes no detectable variance, while task heterogeneity and residual noise dominate.

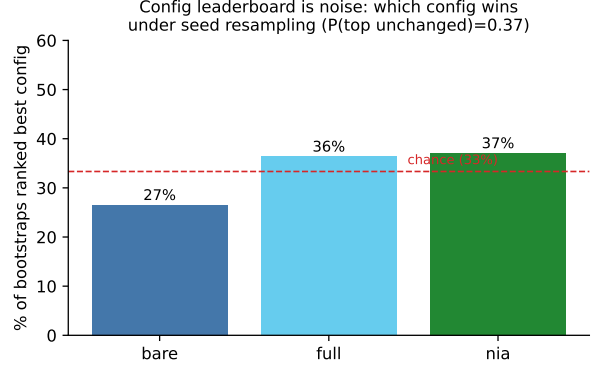


Figure 4: The configuration leaderboard is unstable under resampling. Across 10,000 task-and-seed-resampled leaderboards, each configuration wins the top rank near the 33% chance rate (red line); the observed best configuration stays best only 37% of the time. The tool-augmented configurations (`nia`, `full`) top the board somewhat more than `bare`, a weak, non-significant tendency.

Artifact-order sensitivity. Presenting the six artifact sections in a canonical versus a `stderr`-first order flips 2/20 items (10%), both on T2.1 and both stricter under the permuted order. Because the canonical order is fixed across every run, this does not confound configuration comparisons, but it means absolute scores are conditional on a presentation convention—an within-artifact extension of the candidate-order bias of Wang et al. (2023).

Cross-judge calibration: a strictness offset on documentation criteria. Two same-family judges (`gpt-4o-mini` and `gpt-4o`) scored the same 189 items under modal-of-3. Overall agreement is 82.5% (Cohen’s $\kappa = 0.57$ (Cohen, 1960)), but stratifying by the pre-registered criterion-behavior class is decisive (Figure 5, Table 4): on *capability* criteria (agent can or cannot do the task) the judges agree almost perfectly ($\kappa_a = 0.87$, 94% agreement), while on *documentation/justification* criteria (did the agent adequately justify a contested choice) chance-corrected agreement is exactly zero ($\kappa_c = 0.00$, 39% raw agreement, 14 of 23 items disagreeing). We disclose the mechanism, because it is informative rather than an artifact to hide: on the (c) items `gpt-4o` passes all 23 while `gpt-4o-mini` passes only 9, so one judge is a *constant* rater and Cohen’s κ is mathematically forced to zero regardless of pattern. The finding is therefore not symmetric “noise” but a *directional strictness offset*—`gpt-4o-mini` is uniformly harsher on documentation, never the reverse (the 14 disagreements are all mini-fail / `gpt-4o`-pass). This is not the high-agreement κ -paradox (raw agreement is genuinely low at 39%); paradox-sensitivity coefficients (PABAK -0.22 , Gwet’s AC1 -0.06 ; context only, not κ substitutes (Gwet, 2008; Vach and Gerke, 2023)) confirm the divergence is real. The (c) bucket is small ($n = 23$) and concentrated on the two contested-decision tasks that carry documentation criteria (T2.4, T3.4), so we present this as a criterion-type warning demonstrated on this suite, not a population estimate. Which judge is closer to correct is an interpretive question we take up with human labels below.

The capability-to-documentation drop replicates across model families. To test whether the same-family offset is a GPT-family idiosyncrasy, a cross-family Llama-3.3-70B judge indepen-

Table 3: The three human-adjudicable documentation (c) disagreements, verbatim from the spot-check. Item 10 is a clean case of the stricter judge being wrong; items 6 and 9 (same criterion, two configs) are cases where the criterion text itself conflates the outcome with its justification.

Criterion (item)	mini	gpt-4o	human	Note
T2.4.6 missing-data handled explicitly (item 10)	FAIL	PASS	PASS	Agent ran labeled complete-case analysis (“Complete-case analysis: 474 patients”); gpt-4o-mini over-strict, human sides with gpt-4o .
T2.4.4 exclude post-treatment mediators/colliders (items 6, 9)	FAIL	PASS	AMB	Agents in fact exclude the variables but state no causal rationale; the criterion conflates the outcome (variables absent) with the justification, so neither reading is indefensible.

dently scored the same items (186 of 189 parsed cleanly; 3 dropped for unparseable output). What replicates is the *drop*, not the literal zero: capability agreement stays high ($\kappa_a = 0.87$) while documentation agreement falls to $\kappa_c = 0.23$ (“fair”), with overall agreement lower still ($\kappa = 0.47$). Llama is not a constant rater here, so its $\kappa_c = 0.23$ is a genuine low-agreement value rather than a degenerate zero, which makes the replication of the *pattern* (documentation $\ll$ capability) the more robust claim. The identical $\kappa_a = 0.87$ across families is not shared data—each judge’s scores were computed independently—but reflects the near-unanimity of capability items (all three raters agree on 16 of the 17). Documentation criteria are thus the least reproducible criterion type across judges, within and across families; that is a criterion-type property, not a single-model quirk. Per-task κ (Table 5) is a coarser view because tasks mix criterion types, but it is consistent: T2.4, whose rubric is dominated by two documentation criteria, has the lowest same-family κ (0.122), while T3.4—which carries only one documentation criterion among many capability checks—is diluted back up to 0.710.

A minimal validity anchor—tentatively favoring the lenient judge. Reliability is not validity (Jacobs and Wallach, 2021; Norman et al., 2026): two judges agreeing does not make either correct, and here the two judges *disagree*, so human labels are needed to say which is closer to right. A domain-trained human scored ten items blind to both judges. On the eight items the human could adjudicate unambiguously, agreement was 7/8 with the stricter **gpt-4o-mini** (87.5%, exact CI [47, 99.7]%) and 8/8 with the more lenient **gpt-4o** (100%, exact CI [63, 100]%), the one clean disagreement had **gpt-4o-mini** over-strict and the human siding with **gpt-4o**. This tentatively suggests the documentation offset is **gpt-4o-mini** being miscalibrated-strict rather than a symmetric measurement failure—but two caveats bound the claim hard: the sample is $n = 10$ and non-random (stratified toward disagreement probes, so the exact intervals are descriptive only), and the two items the human *could not* adjudicate are both the same documentation criterion (T2.4.4)—i.e., agreement is highest exactly where the construct is cleanest and undefined where it is not. We therefore claim only a face-validity floor and a directional hint, not a resolution (Table 3 gives the three disagreements verbatim).

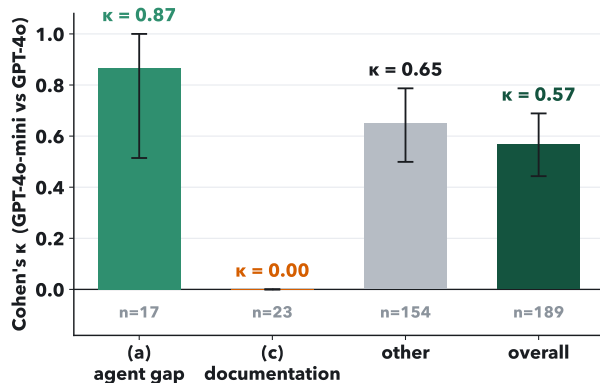


Figure 5: Judges agree on capability, diverge on documentation. Cross-judge Cohen’s κ by pre-registered criterion class: capability criteria (a) reach $\kappa = 0.87$; documentation/justification criteria (c) fall to $\kappa = 0.00$ (a strictness offset—one judge passes all such items—forces κ to zero) despite the same judges and protocol.

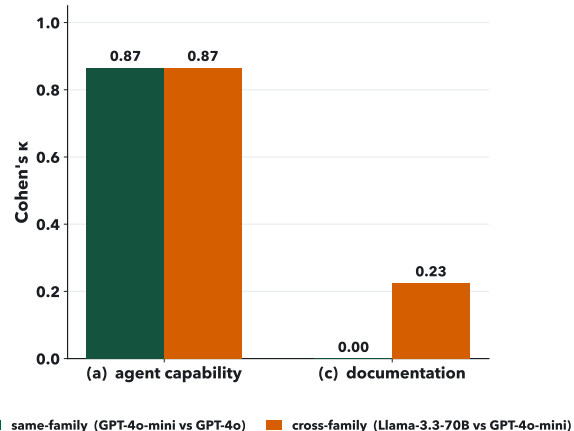


Figure 6: The capability-to-documentation drop is cross-family. A Llama-3.3-70B judge independently reproduces the pattern of the GPT-family judges—high agreement on capability criteria, much lower on documentation—so the divergence is a criterion-type property, not a single-model quirk.

Table 4: Cross-judge agreement drops sharply on documentation criteria, and the drop replicates across families. Cohen’s κ (raw agreement) by criterion class. Same-family: gpt-4o-mini vs. gpt-4o (189 items). Cross-family: gpt-4o-mini vs. Llama-3.3-70B (186 items; 3 unparseable dropped). Same-family $\kappa_c = 0.00$ is a constant-rater artifact (gpt-4o passes all 23 (c) items; see text); the honest quantity is the 39% raw agreement. The n column is the same-family item count per class; cross-family counts differ by at most two per class. Class counts sum to 194 > 189 because five mixed (a)/(c) criterion items are counted in both classes; the overall row is over the 189 distinct items.

Criterion class	n	Same-family κ	Cross-family κ
(a) capability	17	0.87 (94%)	0.87 (94%)
(c) documentation	23	0.00 (39%)	0.23 (55%)
other	154	0.65 (88%)	0.38 (82%)
overall	189	0.57 (83%)	0.47 (79%)

6 A Reliability Protocol for Agent Benchmarks

The practical lesson generalizes beyond this benchmark: decompose before you rank. Our findings motivate a lightweight discipline that any agent-benchmark builder can apply before publishing configuration or model comparisons, extending benchmark-quality checklists (Reuel et al., 2024; Zhu et al., 2025) with a measurement-reliability step they currently lack.

1. **Pre-register the judge protocol and analysis.** Lock scoring thresholds, the aggregation rule (e.g., modal-of- k), and disposition rules before inspecting scores.
2. **Decompose variance by facet.** Fit a G-theory (or crossed random-effects) model over {task, factor-of-interest, seed, judge} and report each variance component with an uncertainty interval—not a clamped point.
3. **Compute the reliability coefficient for the factor you rank on.** Report $E\rho^2$ (or ICC) for the object of comparison against interpretation bands (Koo and Li, 2016; Landis and Koch,

Table 5: Cross-judge agreement by task. Cohen’s κ (raw agreement) per task, for the same-family (gpt-4o-mini vs. gpt-4o) and cross-family (gpt-4o-mini vs. Llama-3.3-70B) judge pairs. T2.4, whose rubric is dominated by documentation criteria, has the lowest same-family κ ; tasks mixing criterion types (e.g. T3.4) are diluted higher, so criterion type rather than task identity drives the class-level pattern.

Task	n	Same-family κ	Cross-family κ
T1.1	18	1.00 (100%)	1.00
T1.4	25	0.865 (96%)	0.865
T2.1	56	0.565 (79%)	0.191
T2.4	36	0.122 (72%)	0.236
T3.1	18	0.211 (72%)	0.211
T3.4	36	0.710 (86%)	0.878

1977). If it is unidentified or poor, *do not publish a ranking* on that factor; report the noise floor instead.

4. **Stress-test the ranking.** Bootstrap the leaderboard and report the probability the top entry changes; a near-chance value is a red flag (Huang et al., 2025; Boubdir et al., 2024; Patel et al., 2026).
5. **Cross-validate the judge and anchor to humans.** Report per-criterion κ across at least two judge families and a small human validity check; disaggregate by criterion type, because aggregate agreement hides criterion-type divergences (and check for constant-rater degeneracy before reading κ).

Applied here, steps 2–4 turn “configurations do not differ significantly” into the sharper, more useful “this benchmark cannot reliably rank configurations at this scale,” and step 5 localizes the largest between-judge divergence to a specific, replicable criterion type.

7 Discussion and Limitations

Our scope is deliberately narrow—depth over breadth—and several limits bound the claims. The study covers one agent model family (Claude Sonnet), one primary judge family, six tasks selected from a larger registry, and $n = 7$ seeds; the cross-family judge replication addresses the judge axis but not the agent axis, and we make no claim that specific numbers transfer to other suites. The task inputs draw on public datasets that may be familiar to the agent from pretraining, but this bears only on absolute accuracy, not on our variance-and-reliability claims, which compare configurations *within* each fixed task and so are unaffected by any shared task familiarity. The value is the *method* and the qualitative instrument findings, which we expect to generalize more readily than the point estimates.

Implications for bioinformatics agent evaluation. The stakes are specific to our field. Computational-biology benchmarks increasingly report single-run agent scores to argue that one tool stack, retrieval setup, or model is better for tasks like differential expression or variant prioritization—the kind of comparison that can steer method adoption. Our results say that at the scale such benchmarks typically run, a single-seed difference is not evidence: the seed noise floor is large enough that the leaderboard reshuffles under resampling. The remedy is cheap and does not require new agents—run a handful of seeds per cell, report the score distribution and a reliability coefficient, and withhold a ranking when that coefficient is not identified. We suggest this become routine for bioinformatics agent papers, alongside the biological-validity checks the community already expects.

The unidentifiability of $E\rho_{\text{config}}^2$ is a direct consequence of small n and only three configuration levels; we report it honestly as unidentifiability rather than as evidence of no effect, and the wide Bayesian interval is itself the finding. A formal equivalence claim is available only for implausibly large effects. The (c) human-validity sample is small ($n = 10$) and T2.4-dominated; we therefore anchor rather than adjudicate, and note that a principled debiasing of judge scores with a *random* human-labeled sample via prediction-powered inference (Angelopoulos et al., 2023) is the natural next step—our labels were stratified toward disagreement probes and so do not satisfy its sampling assumption. Reproducibility carries two documented caveats: the judge is specified by alias (snapshot recorded post hoc, not pinned) and per-call `system_fingerprint` was not captured on the calibration path. There is an irony worth stating: a paper arguing for measurement rigor characterizes an instrument that is itself imperfectly pinned, which bounds the force of our own reproducibility claims; the pre-registration commits are verifiable only once the (currently private) repository is released, and an anonymized artifact would accompany a camera-ready. Our nearest methodological neighbor, Total Evaluation Error (Messing, 2026), targets annotation pipelines; our contribution is the agent-configuration setting, the documentation-criterion judge divergence with its disclosed strictness mechanism, and its cross-family replication.

8 Conclusion

Treating a bioinformatics agent benchmark as a measuring instrument reveals that **single-seed scores cannot support the comparisons the benchmark is built to make**: seed variance dominates, the configuration reliability coefficient is unidentifiable across three methods (its interval spans nearly $[0, 1]$), and the leaderboard reshuffles under seed resampling—the observed best configuration stays best barely above chance. A distinct, irreducible part of the noise is the LLM judge, whose scoring of documentation/justification criteria diverges sharply between models—a systematic strictness offset that replicates, as a capability-to-documentation drop, across model families. Task difficulty, which the benchmark does not target, is separated more reliably (partly for structural reasons); the instrument is not globally uninformative, it simply cannot resolve the system comparison it was built for at this scale. For LLM coding agents on bioinformatics tasks, reliability—not single-run accuracy—must be measured and reported before a benchmark is used to rank them. We offer a *decompose-before-you-rank* protocol to that end.

Data and code availability. The 126 runs, 189 calibration items, pre-registration commits (with hashes), analysis scripts, and figure-generating code are maintained in a version-controlled project repository and will be released publicly upon publication; every reported number traces to a committed artifact. Task inputs derive from public resources (e.g., Himes et al. 2014 GEO SRP033351, TCGA-LUAD, GIAB NA12878) under their respective licenses.

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